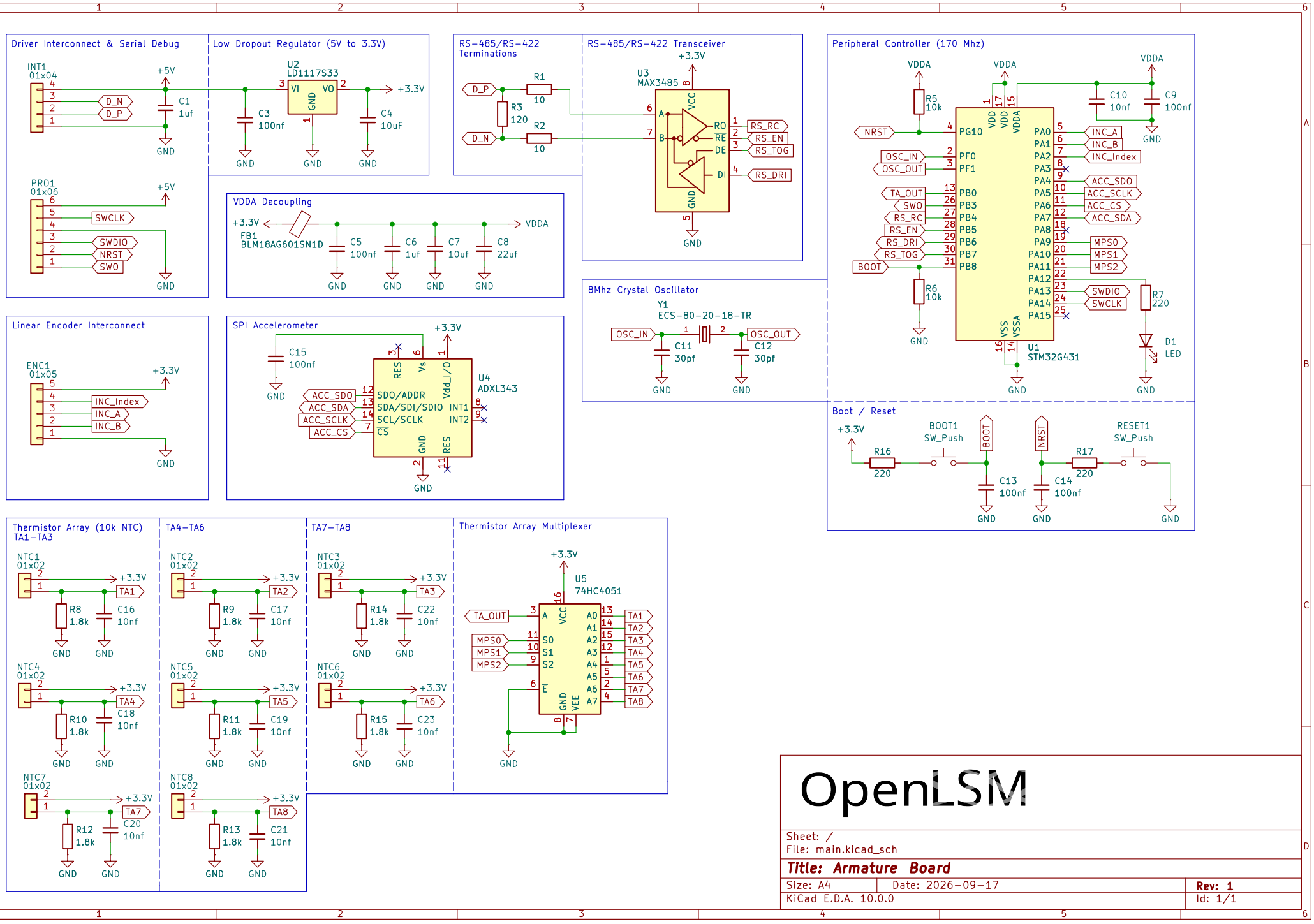

Driver Interconnect & Serial Debug
INT1
01x04
D_N
D_P
+5V
C1
1uf
GND
PRO1
01x06
SWCLK
SWDIO
NRST
SWO
Low Dropout Regulator (5V to 3.3V)
U2
LD1117S33
VI
VO
GND
C3
100nf
C4
10uF
+3.3V
VDDA Decoupling
FB1
BLM18AG601SN1D
C5
100nf
C6
1uf
C7
10uf
C8
22uf
VDDA
RS-485/RS-422
Terminations
R1
10
R2
10
R3
120
RS-485/RS-422 Transceiver
U3
MAX3485
VCC
RO
RE
DE
DI
RS_RC
RS_EN
RS_TOG
RS_DRI
8Mhz Crystal Oscillator
Y1
ECS-80-20-18-TR
OSC_IN
OSC_OUT
C11
30pf
C12
30pf
Peripheral Controller (170 Mhz)
U1
STM32G431
PG10
PF0
PF1
PB0
PB3
PB4
PB5
PB6
PB7
PB8
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
VDD
VDDA
VSS
VSSA
NRST
BOOT
TA_OUT
SWO
INC_A
INC_B
INC_Index
ACC_SDO
ACC_SCLK
ACC_CS
ACC_SDA
MPS0
MPS1
MPS2
SWDIO
SWCLK
R5
10k
R6
10k
R7
220
D1
LED
C9
100nf
C10
10nf
Boot / Reset
BOOT1
SW_Push
RESET1
SW_Push
R16
220
R17
220
C13
100nf
C14
100nf
Linear Encoder Interconnect
ENC1
01x05
INC_Index
INC_A
INC_B
SPI Accelerometer
U4
ADXL343
SDO/ADDR
SDA/SDI/SDIO
SCL/SCLK
CS
Vdd_I/O
INT1
INT2
RES
Vs
C15
100nf
ACC_SDO
ACC_SDA
ACC_SCLK
ACC_CS
Thermistor Array (10k NTC)
TA1-TA3
TA4-TA6
TA7-TA8
NTC1
NTC2
NTC3
NTC4
NTC5
NTC6
NTC7
NTC8
01x02
R8
1.8k
R9
1.8k
R10
1.8k
R11
1.8k
R12
1.8k
R13
1.8k
R14
1.8k
R15
1.8k
C16
10nf
C17
10nf
C18
10nf
C19
10nf
C20
10nf
C21
10nf
C22
10nf
C23
10nf
TA1
TA2
TA3
TA4
TA5
TA6
TA7
TA8
Thermistor Array Multiplexer
U5
74HC4051
TA_OUT
MPS0
MPS1
MPS2
S0
S1
S2
A0
A1
A2
A3
A4
A5
A6
A7
E
VCC
GND
VEE
OpenLSM
Sheet: /
File: main.kicad_sch
Title: Armature Board
Size: A4
Date: 2026-09-17
Rev: 1
KiCad E.D.A. 10.0.0
Id: 1/1